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Step 1: Data source for the Assignment 2

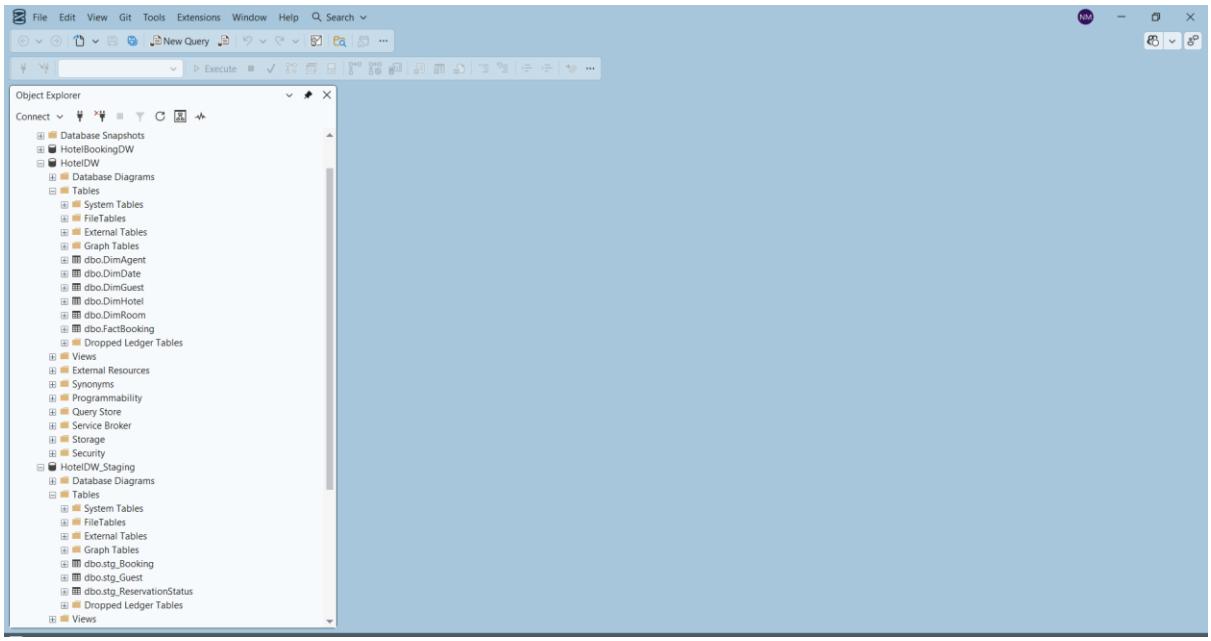
The data warehouse created in Assignment 1 was used as the data source for this assignment. It contains fact and dimension tables organized in a star schema, which supports analytical processing and reporting.

- Database Name : HotelDW
- Schema Type : Star Schema

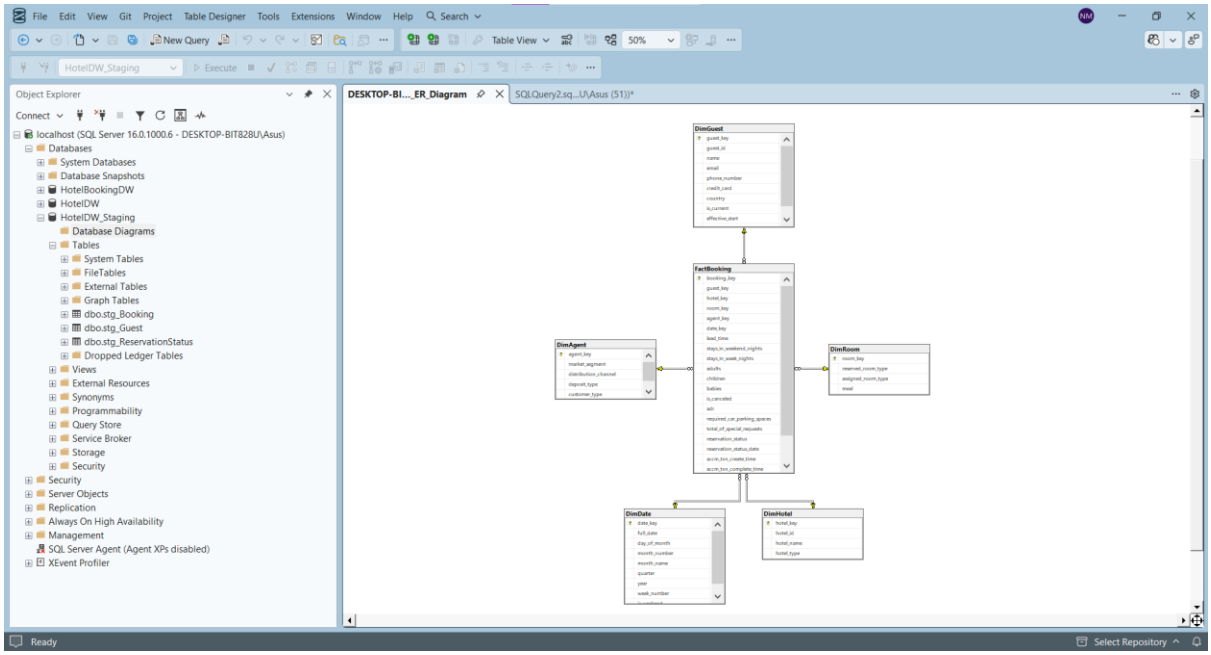
Tables

Table	Type	Description
FactBooking	Fact Table	Hotel booking transactions
DimGuest	Dimension	Guest/Customer details
DimRoom	Dimension	Room details
DimHotel	Dimension	Hotel details
DimAgent	Dimension	Booking agent details
DimDate	Dimension	Date details

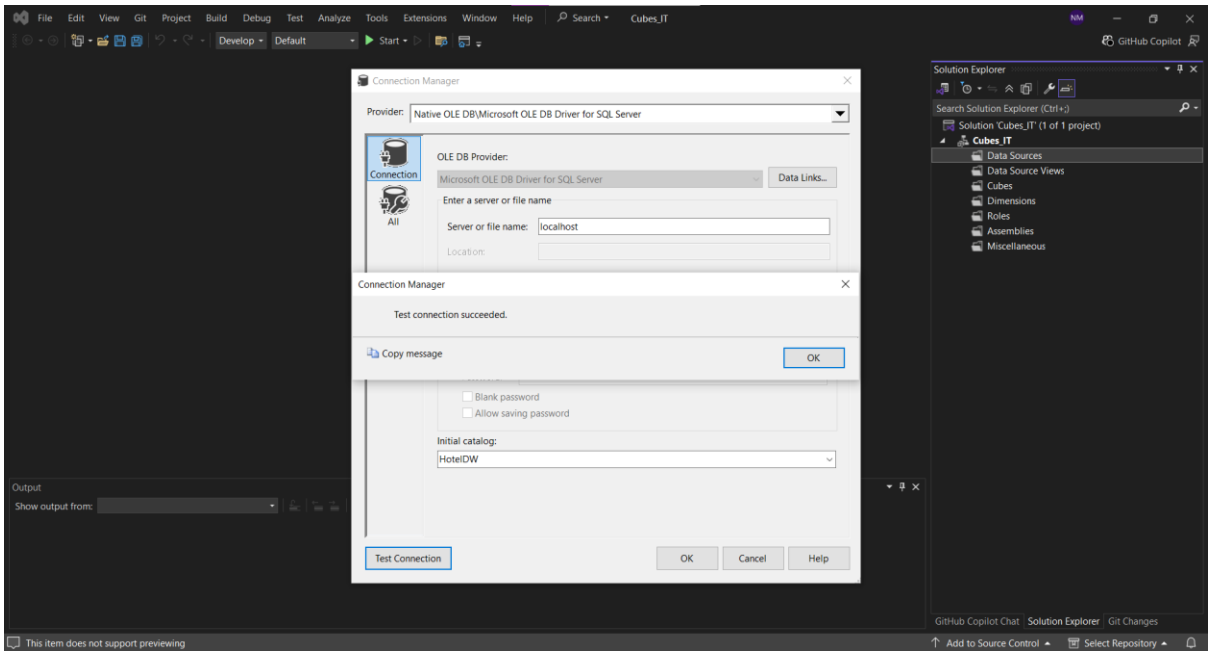
Database



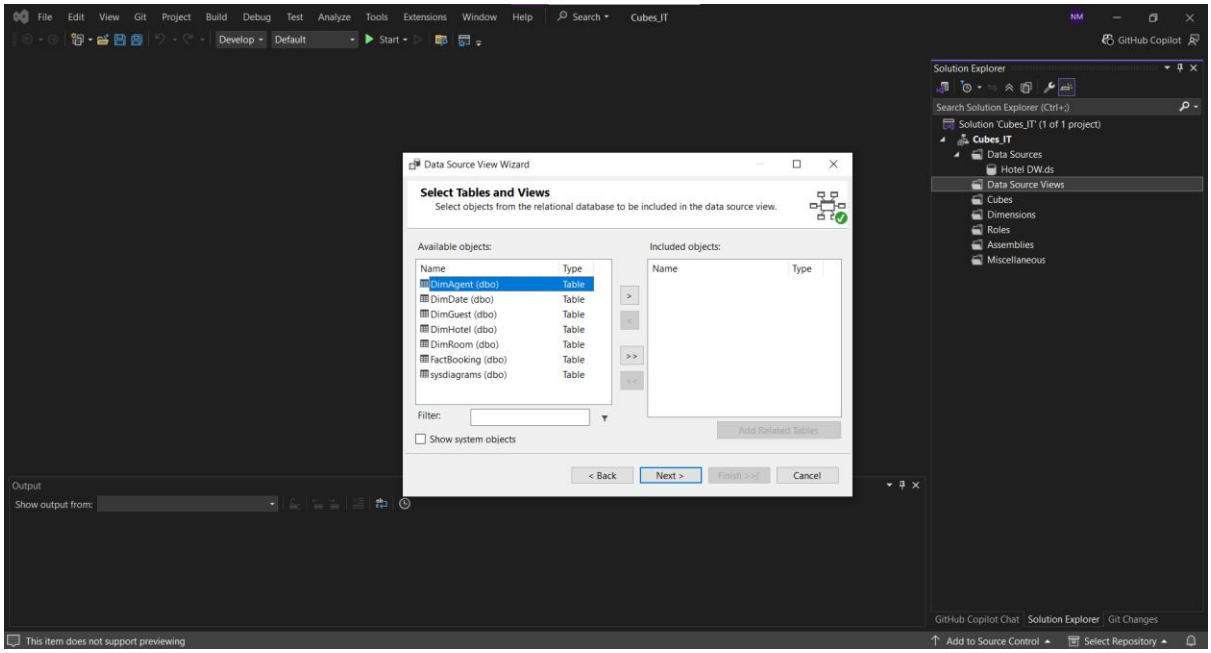
ER Diagram

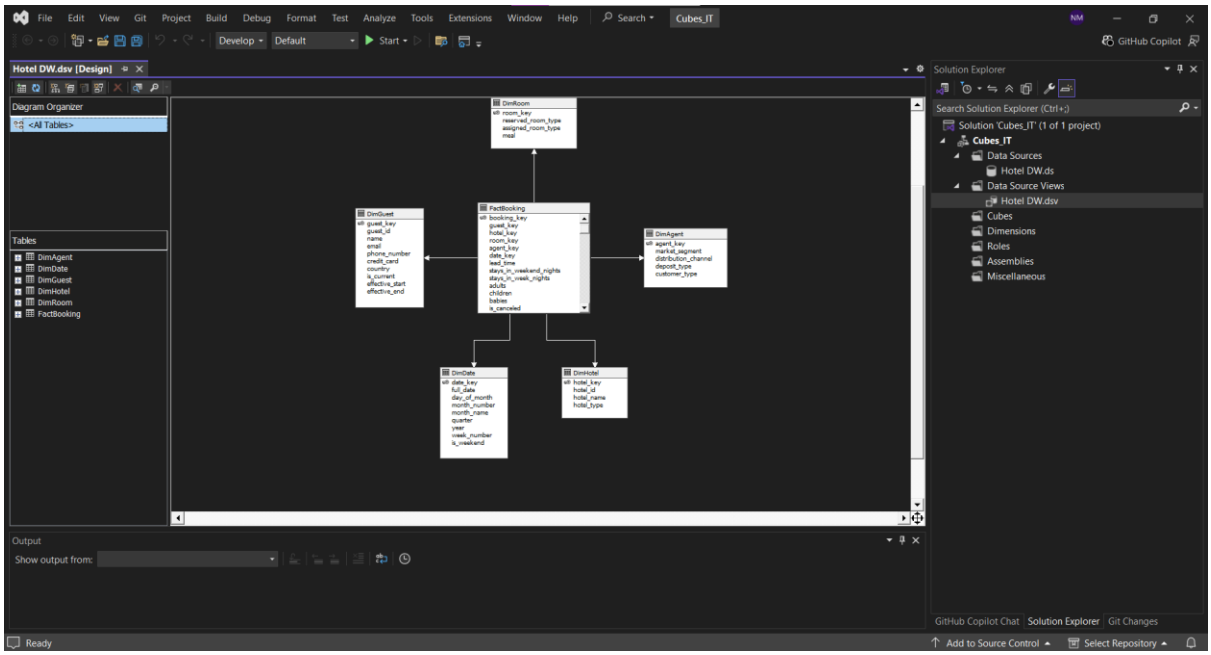


Data Source



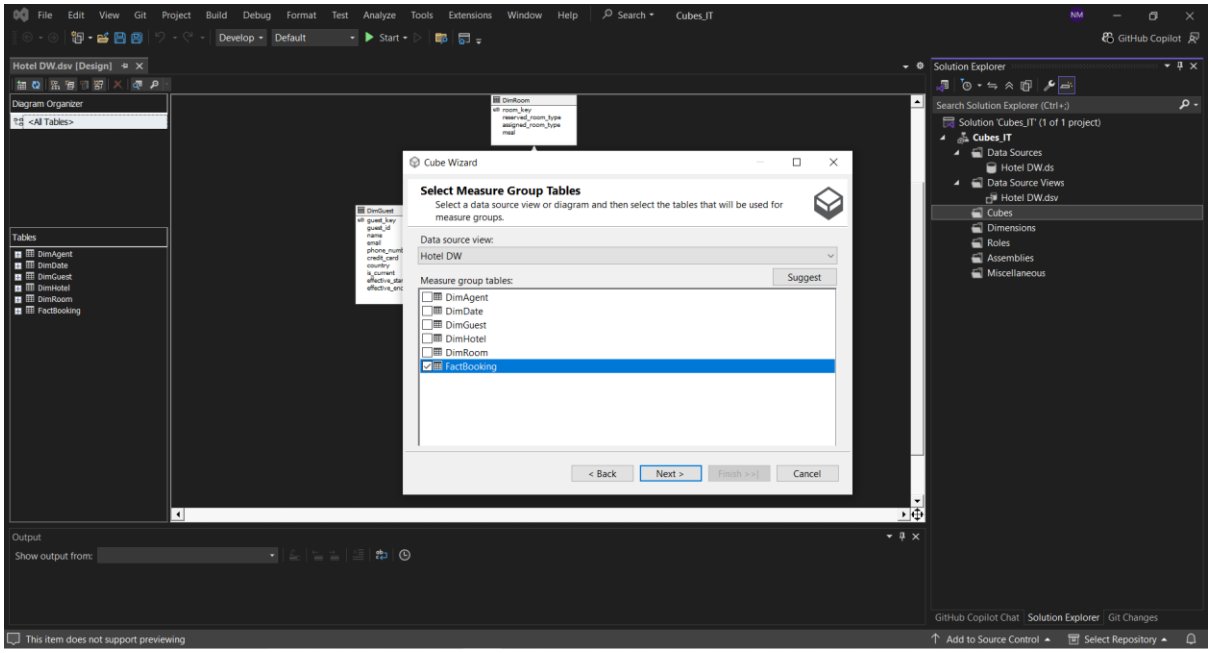
Data Source View

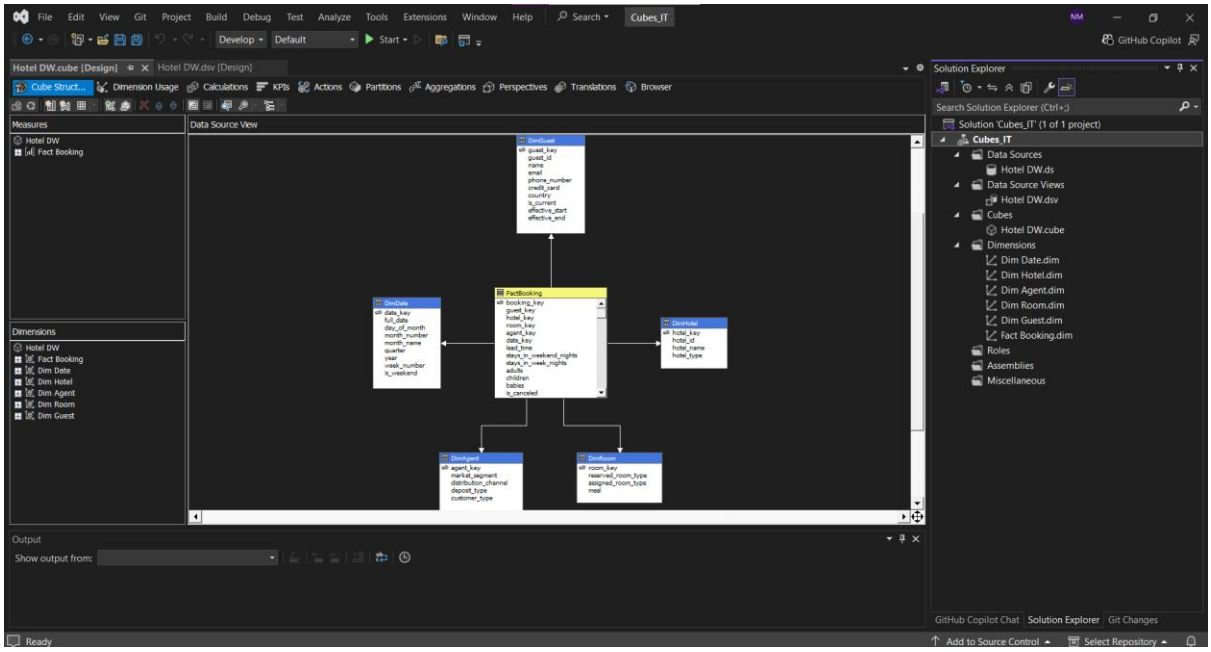


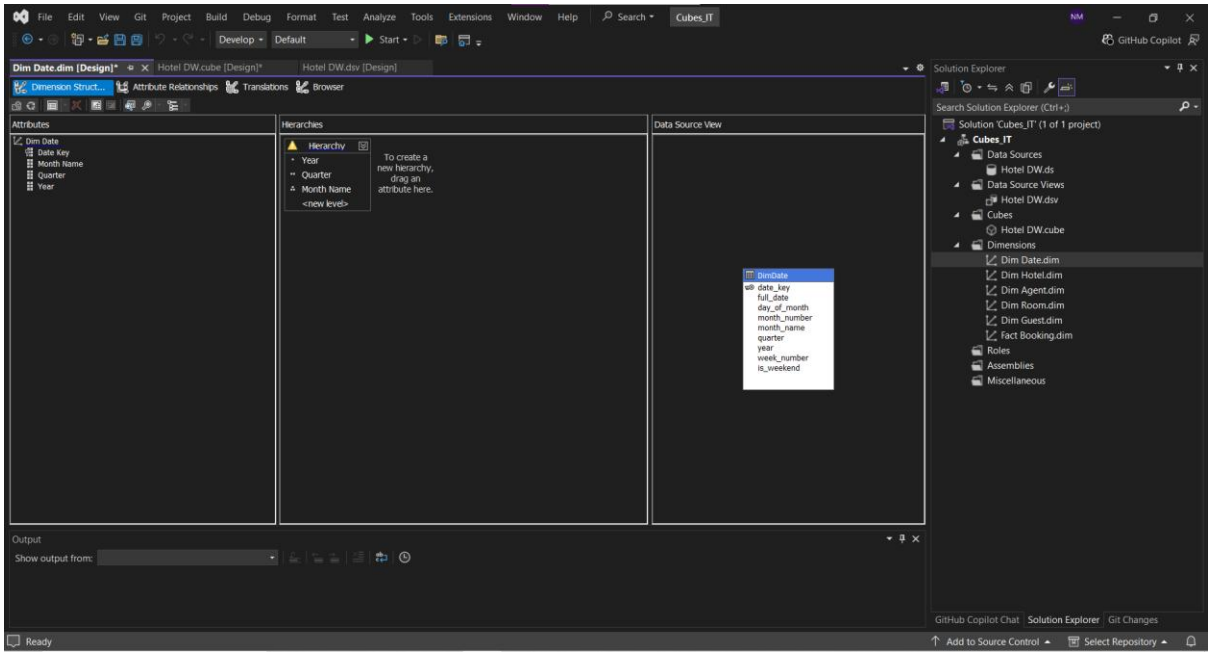


Step 2: SSAS Cube implementation

The SSAS cube was created using the FactBooking table as the measure group. Dimension tables including DimDate, DimHotel, DimRoom, DimAgent, and DimGuest were connected to the cube using their respective foreign keys. A hierarchy was implemented in the DimDate dimension using Year → Quarter → Month levels.



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Cubes_IT
Develop
Default
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GitHub Copilot
Dim Date.dim [Design]*
Hotel DW.cube [Design]*
Hotel DW.dsv [Design]
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Translations
Browser
Attributes
Dim Date
Date Key
Month Name
Quarter
Year
Hierarchies
Hierarchy
Year
Quarter
Month Name
<new level>
To create a new hierarchy, drag an attribute here.
Data Source View
DimDate
date_key
full_date
day_of_month
month_number
month_name
quarter
year
week_number
is_weekend
Output
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Solution Explorer
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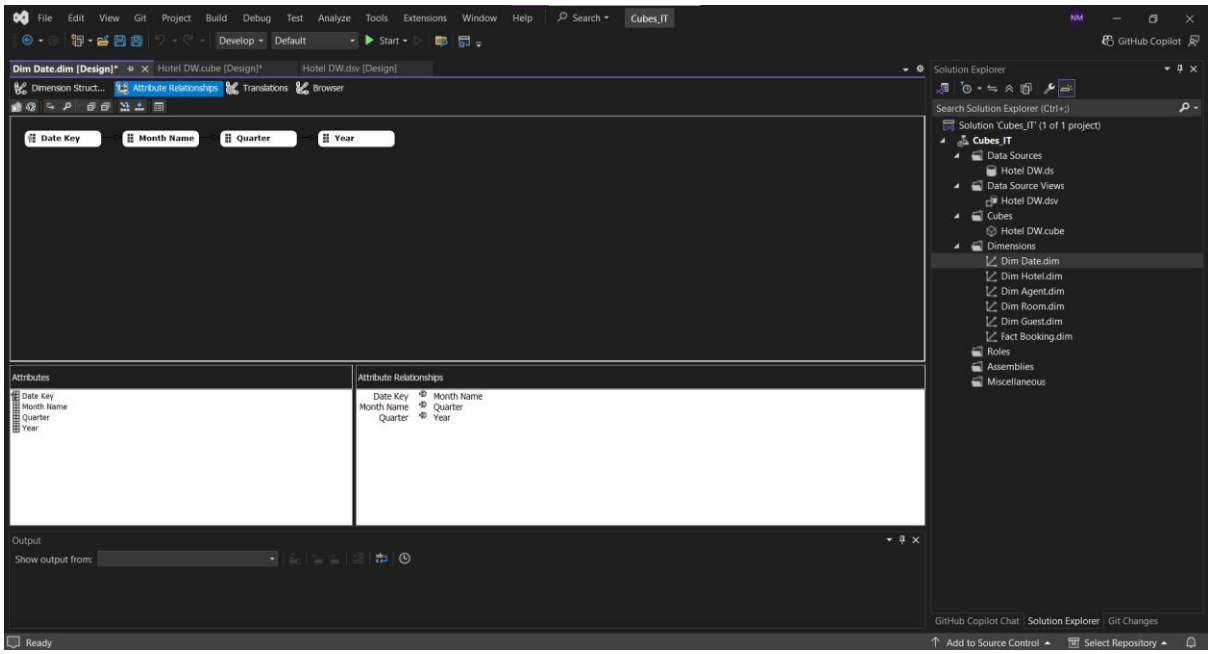
Solution 'Cubes_IT' (1 of 1 project)
Cubes_IT
Data Sources
Hotel DW.ds
Data Source Views
Hotel DW.dsv
Cubes
Hotel DW.cube
Dimensions
Dim Date.dim
Dim Hotel.dim
Dim Agent.dim
Dim Room.dim
Dim Guest.dim
Fact Booking.dim
Roles
Assemblies
Miscellaneous
GitHub Copilot Chat
Solution Explorer
Git Changes
Ready
Add to Source Control
Select Repository


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search
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Hotel DW.dsv [Design]
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Attribute Relationships
Translations
Browser
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Month Name
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Output
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Dim Agent.dim
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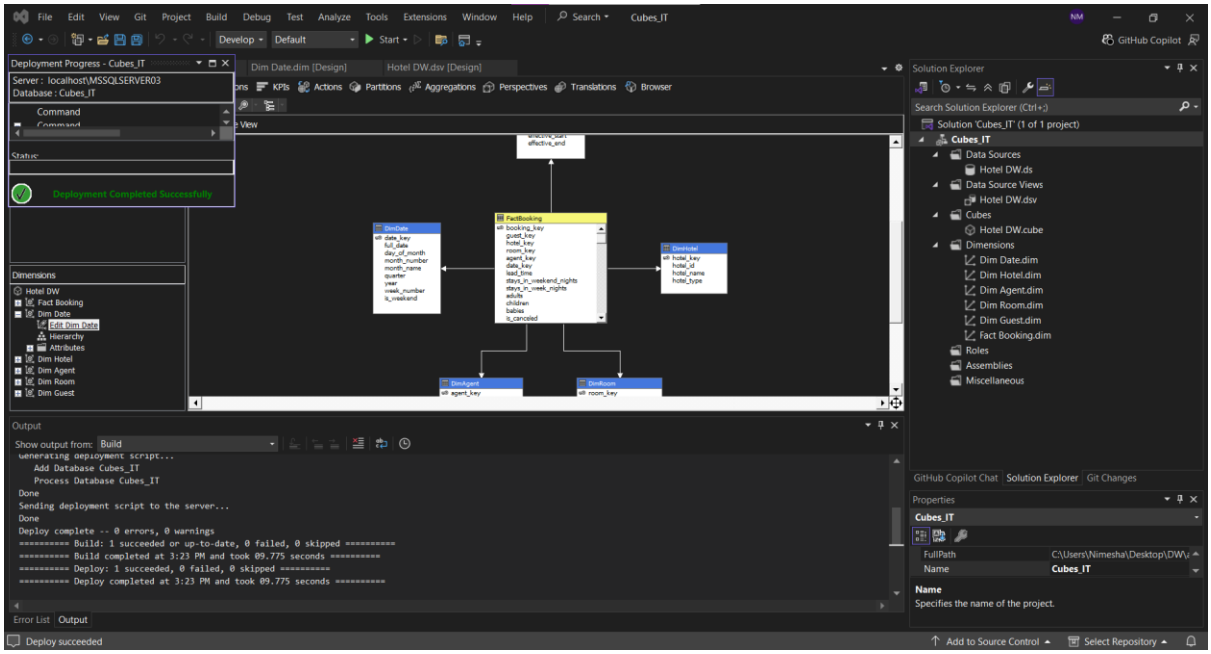
Select Repository

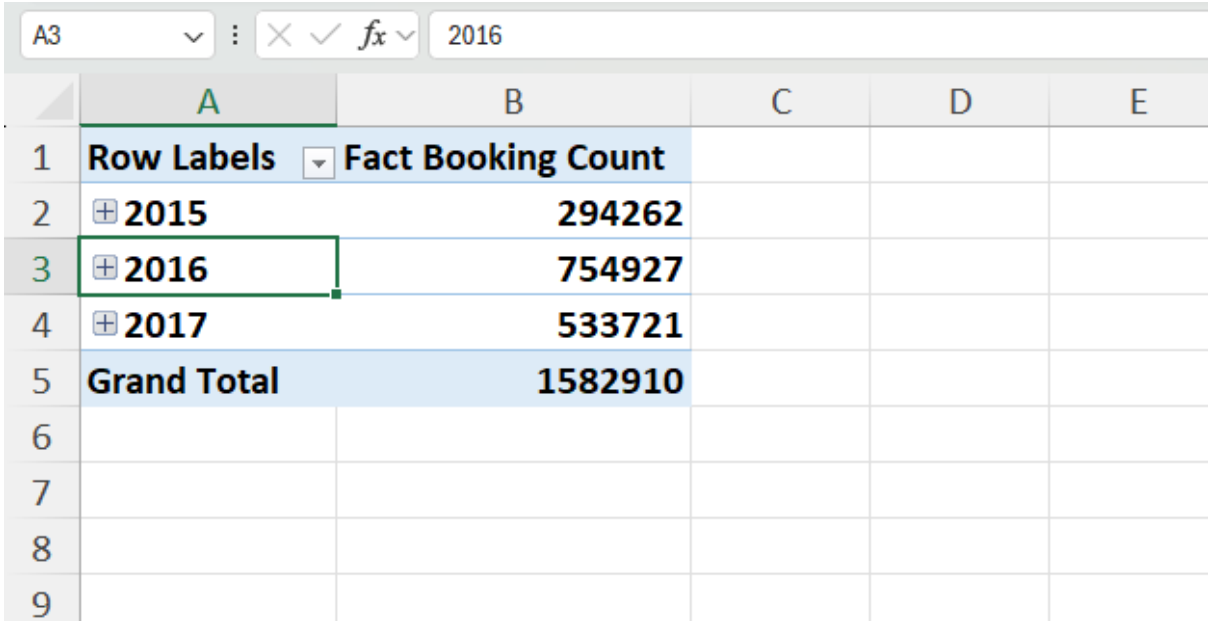


Step 3: Demonstration of OLAP operations

In this section, the Hotel DW OLAP cube deployed on SQL Server Analysis Services (SSAS) was connected to Microsoft Excel using the Data tab. A PivotTable was created by connecting to the local Analysis Services instance (localhost\MSSQLSERVER03) and selecting the Cubes_IT database with the Hotel DW cube. The following five OLAP operations were demonstrated using PivotTable configurations.

3.1 Roll – Up

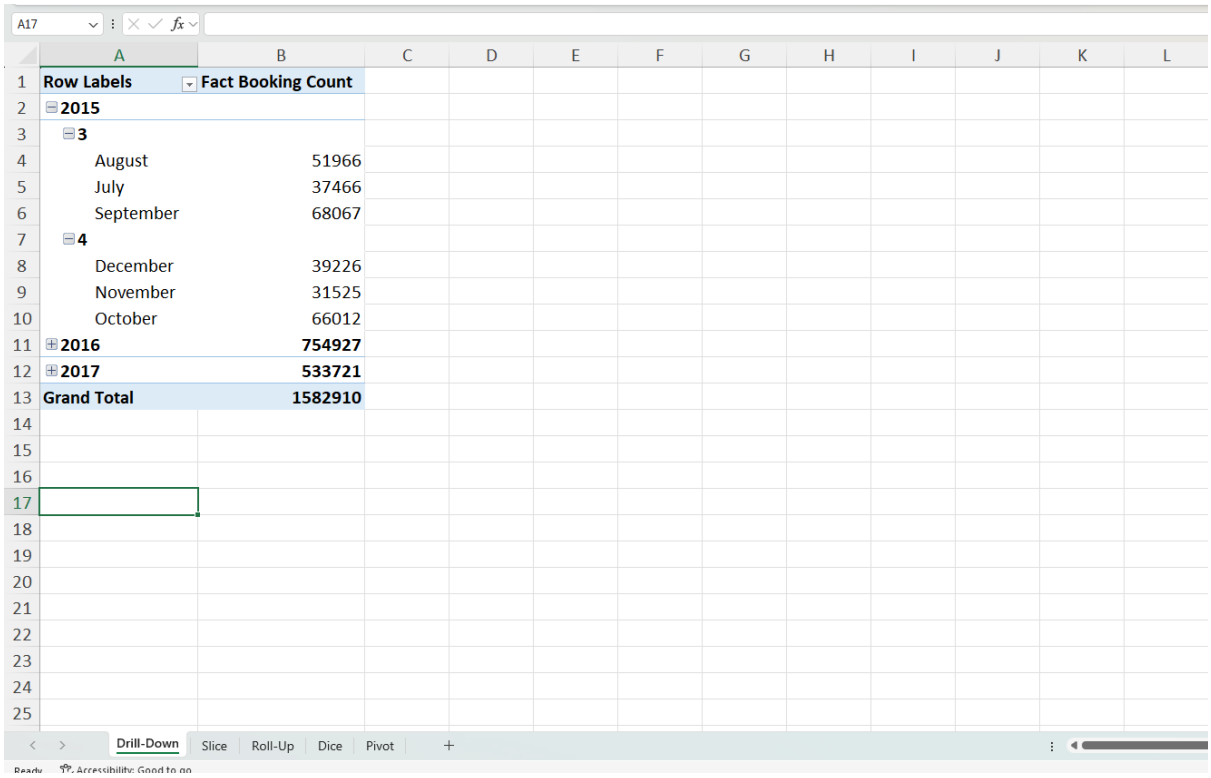
The data was aggregated from a detailed level to a higher summary level. The Year field from the Dim Date dimension was added to the Rows area and Fact Booking Count was added to the Values area, summarizing total bookings per year.



Row Labels	Fact Booking Count
2015	294262
2016	754927
2017	533721
Grand Total	1582910

3.2 Drill – Down

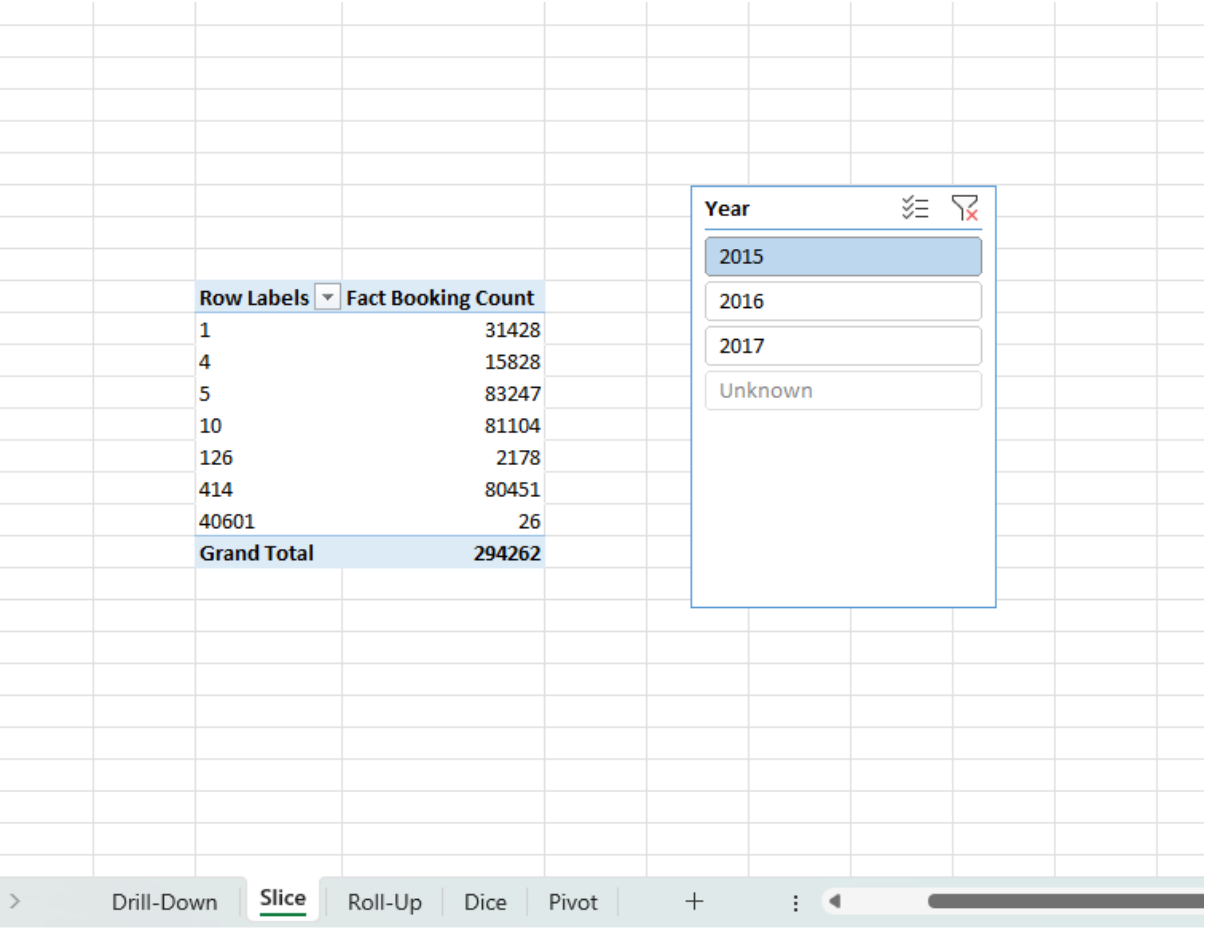
Starting from the Year-level summary, the Quarter field was added to the Rows area beneath Year. This expanded the data to show booking counts broken down by quarter within each year, moving from a high-level summary to a more detailed view.



Row Labels	Fact Booking Count
2015	
3	
August	51966
July	37466
September	68067
4	
December	39226
November	31525
October	66012
2016	754927
2017	533721
Grand Total	1582910

3.3 Slice

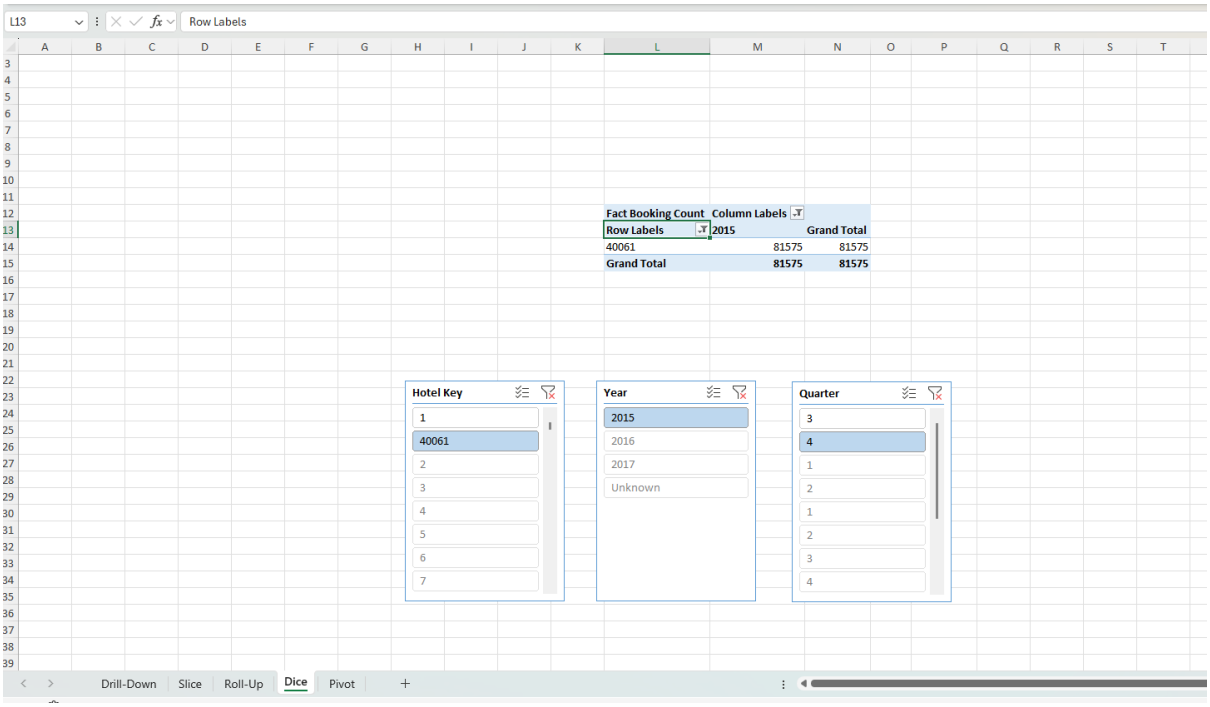
A single dimension filter was applied by dragging the Year field to the Filters area and selecting only the year 2015. This isolated a single "slice" of the data cube, displaying only bookings for 2015 .



Row Labels	Fact Booking Count
1	31428
4	15828
5	83247
10	81104
126	2178
414	80451
40601	26
Grand Total	**294262**

3.4 Dice

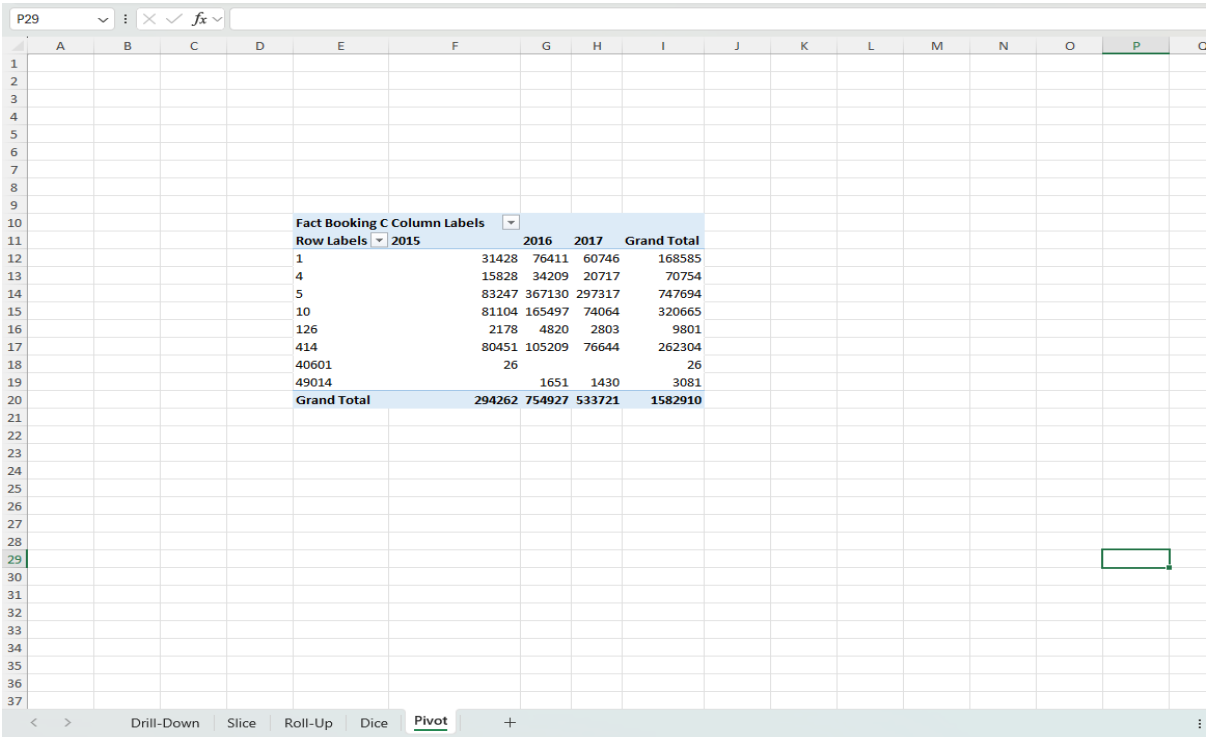
Two dimension filters were applied simultaneously . Year was set to 2015 and Quarter was added as a second filter. This operation selected a sub-cube by filtering across multiple dimensions at once, narrowing the data to a specific intersection.



Fact Booking Count	Column Labels	
Row Labels	2015	Grand Total
40061	81575	81575
Grand Total	81575	81575

3.5 Pivot

The Year field was moved from the Rows area to the Columns area. This rotated the data perspective so that years appeared as column headers (2015, 2016, 2017) rather than row labels, demonstrating how the same data can be viewed from a different axis.

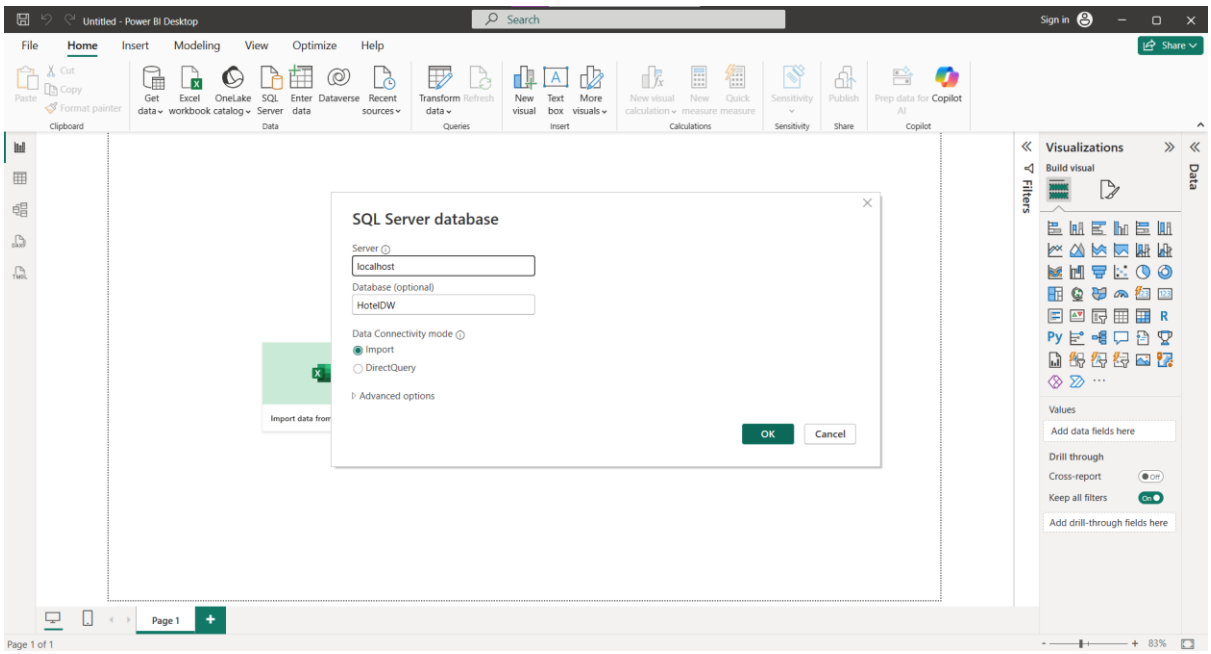


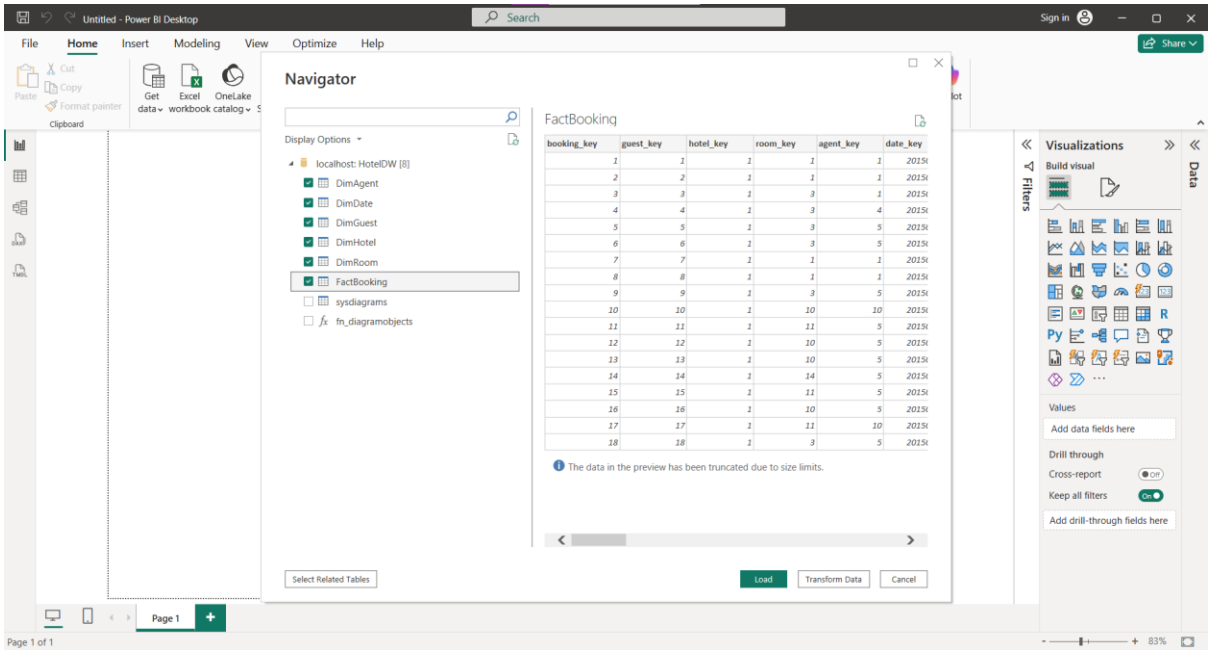
Fact Booking C	Column Labels			
Row Labels	2015	2016	2017	Grand Total
1	31428	76411	60746	168585
4	15828	34209	20717	70754
5	83247	367130	297317	747694
10	81104	165497	74064	320665
126	2178	4820	2803	9801
414	80451	105209	76644	262304
40601	26			26
49014		1651	1430	3081
Grand Total	294262	754927	533721	1582910

Step 4: PowerBI Reports

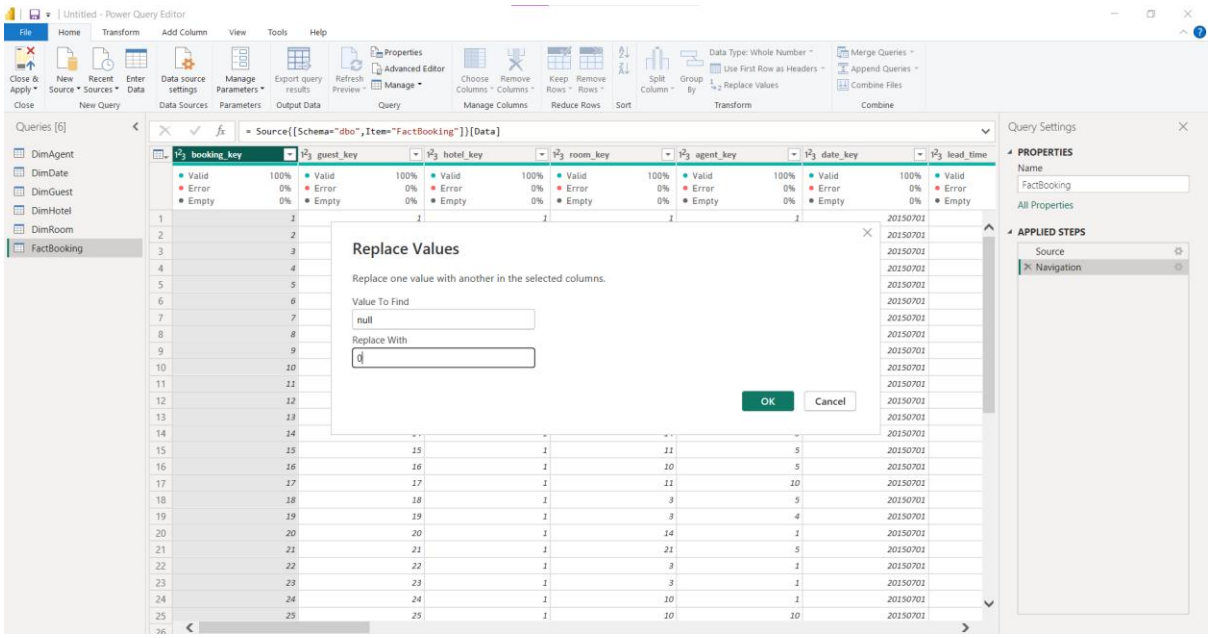
The data warehouse developed in Assignment 1 was utilized as the primary data source for building interactive business intelligence reports using Microsoft Power BI. The objective of this step is to transform structured warehouse data into meaningful visual insights that support decision-making in the hotel booking domain.

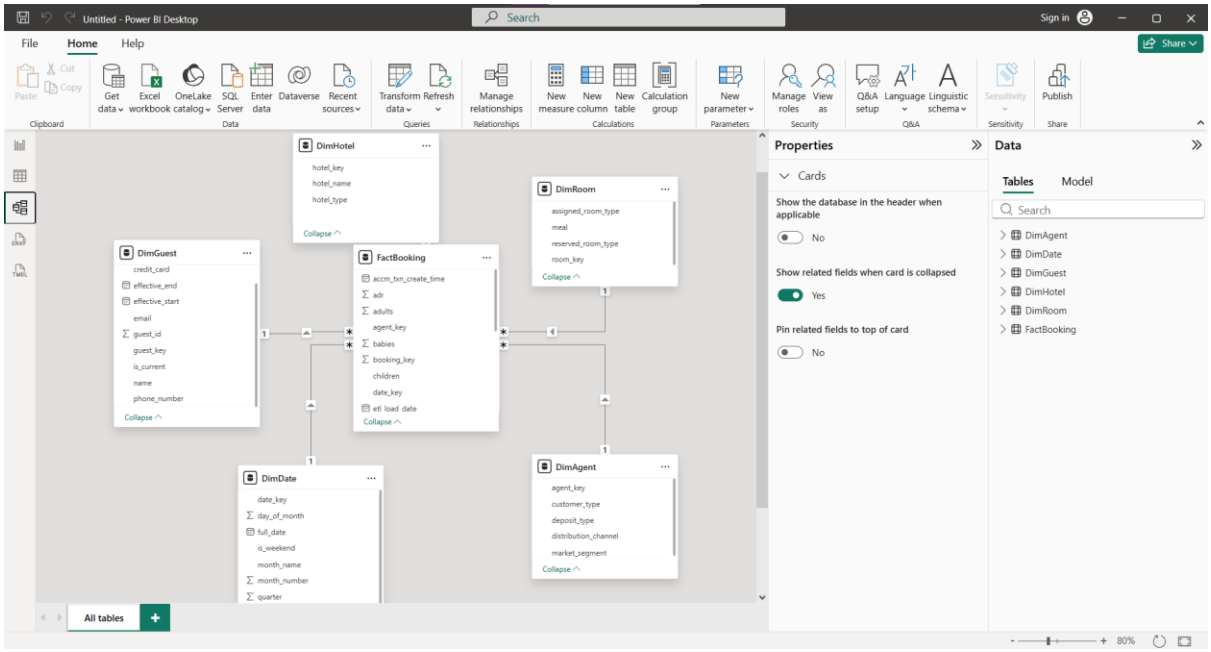
The Power BI reports are designed to demonstrate key analytical capabilities such as data summarization, filtering, hierarchical exploration, and detailed drill-down analysis.





Data Preparations





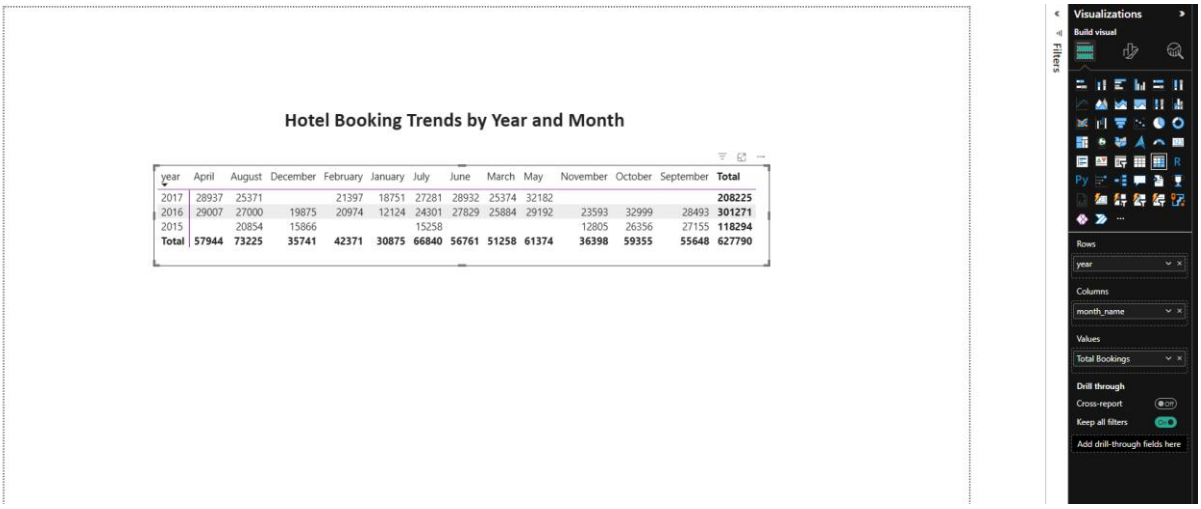
Report 1: Matrix Visual — Hotel Booking Summary

A new page named **"Matrix"** was created in Power BI Desktop. A Matrix visual was inserted from the Visualizations pane. The **Year** field from the DimDate table was placed in the **Rows** section to group data by year. The **Month_Name** field from the DimDate table was placed in the **Columns** section to provide a monthly breakdown of bookings. The **Total Bookings** measure, created using the COUNT of booking_key from the FactBooking table, was used in the **Values** section to display the total number of bookings for each year and month combination.

The Matrix visual displays booking distribution across months for each year, enabling detailed temporal analysis of booking trends. This structure allows comparison of booking patterns across different months within each year. The visual title was set using the Format Visual panel under General settings.

Hotel Booking Trends by Year and Month

year	April	August	December	February	January	July	June	March	May	November	October	September	Total
2017	28937	25371		21397	18751	27281	28932	25374	32182				**208225**
2016	29007	27000	19875	20974	12124	24301	27829	25884	29192	23593	32999	28493	**301271**
2015		20854	15866			15258				12805	26356	27155	**118294**
Total	**57944**	**73225**	**35741**	**42371**	**30875**	**66840**	**56761**	**51258**	**61374**	**36398**	**59355**	**55648**	**627790**

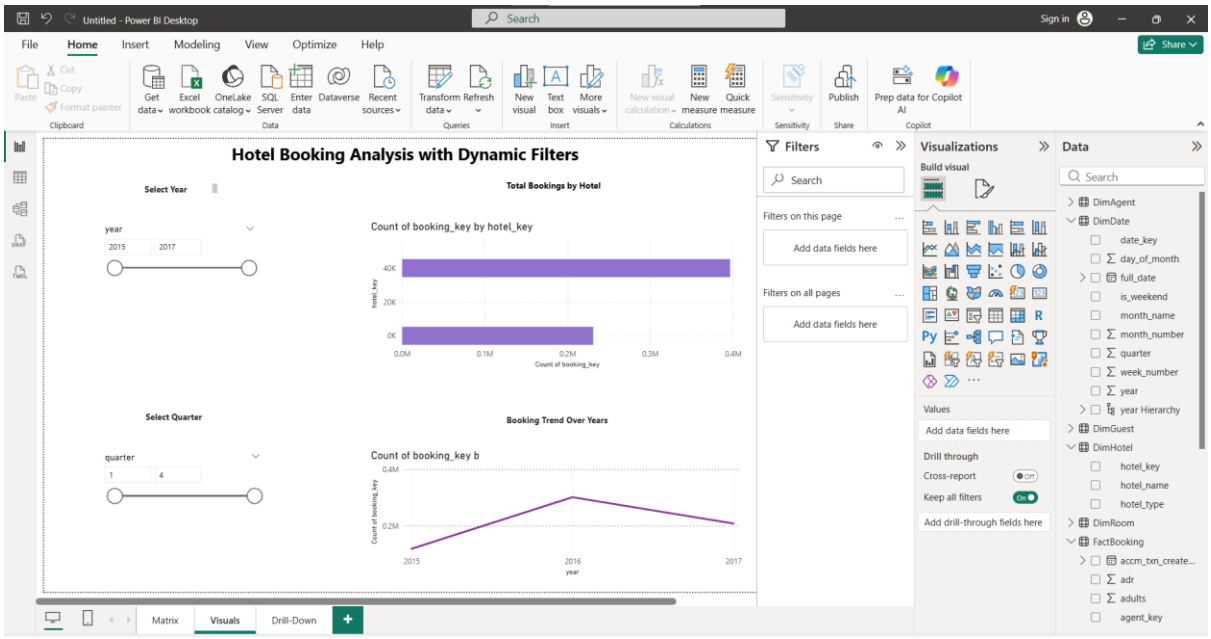


Report 2: Cascading Slicers with Multiple Visuals

A new page named "Visuals" was created. The page title "Hotel Booking Analysis with Dynamic Filters" was added using a Text Box visual. Two Slicer visuals were added to the left side of the canvas. The first slicer was assigned the year field from DimDate with the title "Select Year", and the second slicer was assigned the quarter field from DimDate with the title "Select Quarter". To implement cascading filters, the year slicer was selected, Edit Interactions was enabled from the Format tab, and the filter interaction was set so that selecting a year value dynamically filters the available quarters in the second slicer.

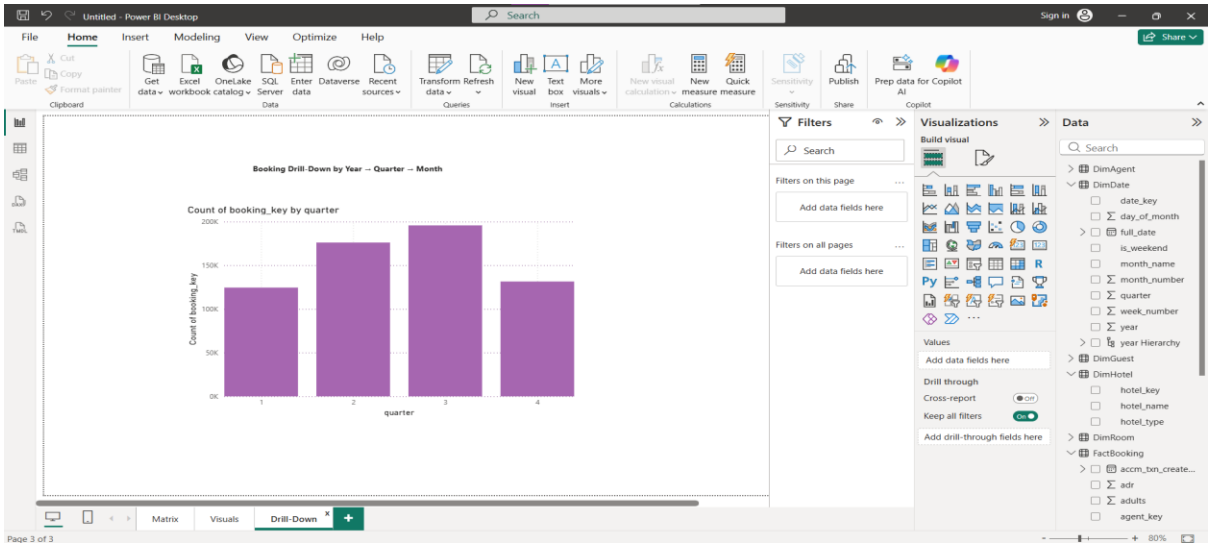
A Clustered Bar Chart titled "Total Bookings by Hotel" was added to the right side of the canvas, with hotel_key on the Y-axis and Count of booking_key on the X-axis. A Line Chart titled "Booking Trend Over Years" was added below it, with year on the X-axis and Count of

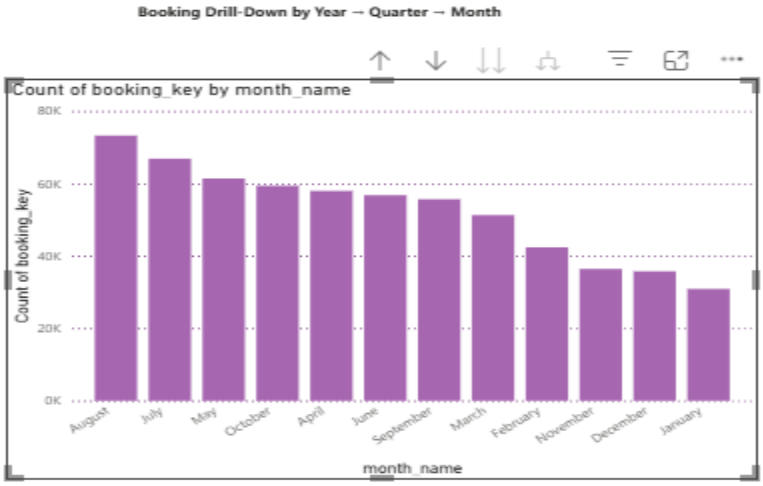
booking_key on the Y-axis. Both visuals respond dynamically when values are selected in either slicer, enabling interactive filtering of the data.

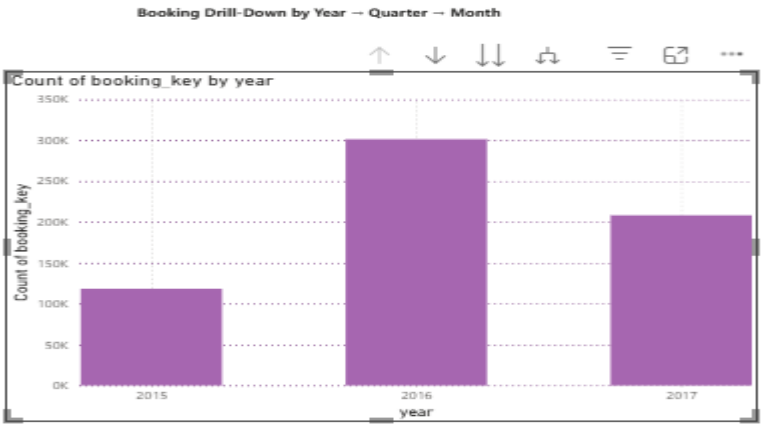


Report 3: Drill-Down Hierarchy Report

A new page named "Drill-Down" was created. A date hierarchy named "year Hierarchy" was built within the DimDate table by right-clicking the year field in the Data pane and selecting Create Hierarchy. The quarter and month_name fields were then added to the hierarchy in order, creating a three-level structure of Year, Quarter, and Month. A Clustered Column Chart was inserted and the year Hierarchy was placed on the X-axis while Count of booking_key was placed on the Y-axis. The visual title was set to "Booking Drill-Down by Year → Quarter → Month". Drill-down mode was enabled by clicking the drill-down arrow icon on the visual. Clicking on a year bar navigates to the quarter level, and clicking again navigates to the month level, allowing users to explore booking data hierarchically.


Untitled - Power BI Desktop
Booking Drill-Down by Year → Quarter → Month
Count of booking_key by quarter
Count of booking_key
quarter
Filters
Visualizations
Data
Matrix
Visuals
Drill-Down

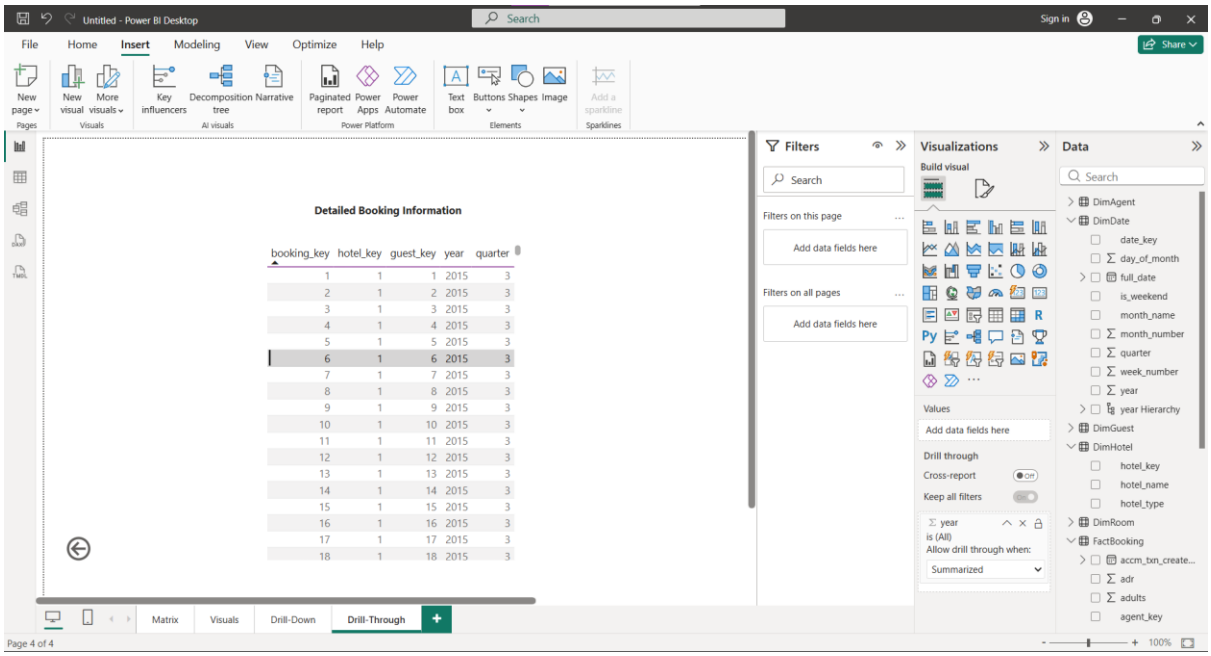

Booking Drill-Down by Year → Quarter → Month
Count of booking_key by month_name
Count of booking_key
80K
60K
40K
20K
0K
August
July
May
October
April
June
September
March
February
November
December
January
month_name


Booking Drill-Down by Year → Quarter → Month
Count of booking_key by year
Count of booking_key
350K
300K
250K
200K
150K
100K
50K
0K
2015
2016
2017
year

Report 4: Drill-Through Report

A new page named "Drill-Through" was created to serve as the detail page. A Table visual was inserted and populated with booking_key, hotel_key, guest_key, year, and quarter fields to display detailed booking records. In the Visualizations pane, the year field was dragged into the Drill through section, enabling this page to receive drill-through context from other report pages. A Back button was inserted using Insert → Buttons → Back, allowing users to return to the previous page after viewing details. The page title was set to "Detailed Booking Information".

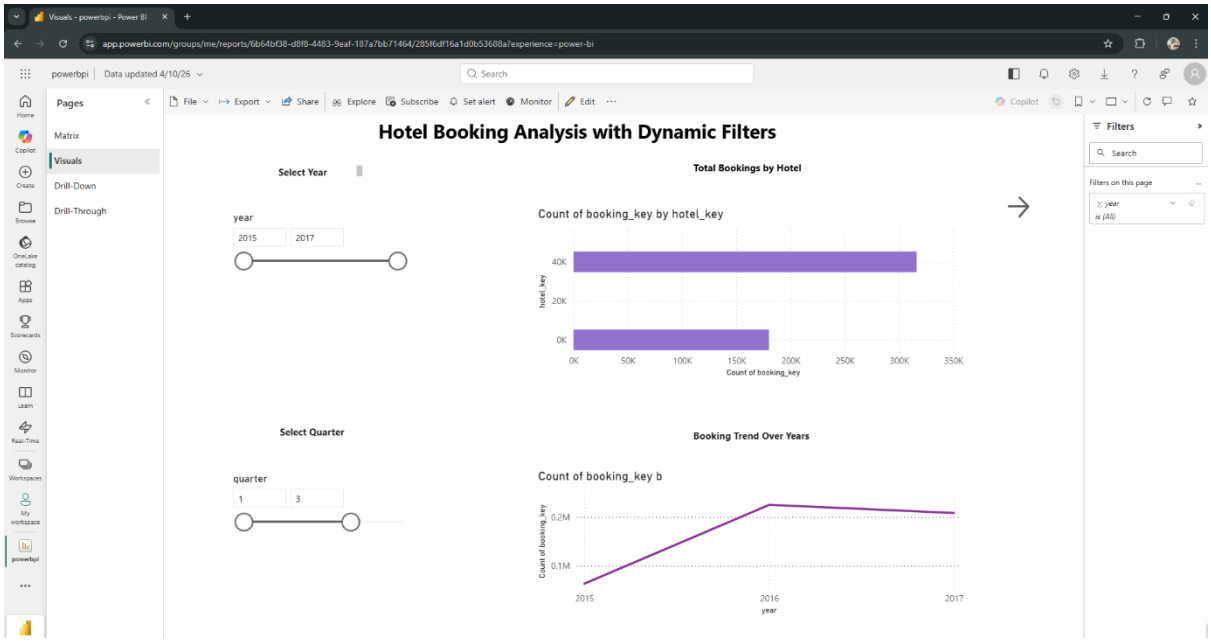
To demonstrate the drill-through functionality, the Visuals page was used as the source. Right-clicking on a data point in the Bar Chart reveals the Drill through option in the context menu, which navigates to the Drill-Through page filtered by the selected year value, displaying only the relevant booking records for that year.



Publishing to Power BI Service

After completing the above four reports in Microsoft Power BI Desktop, the report file was saved and published to the Power BI Service using the publish option in the Home tab. After signing in with my Microsoft account, the appropriate workspace was selected and the report was successfully uploaded.

The published reports were then accessed through the Power BI Service (app.powerbi.com) to verify that all visuals, slicers, drill-down, and drill-through functionalities worked correctly in the online environment.



Conclusion

In this assignment, a complete BI solution was implemented starting from a data warehouse to SSAS cube, Excel OLAP analysis, and Power BI reporting. The system enables efficient data analysis and supports decision-making through interactive visualizations.